# Reproducibility Report for Study by Salomons et al. (2026, Animal Behaviour)

Samantha Kefer (skefer@ucsd.edu)

2026-10-25

## Table of contents

Introduction . . . 1
Justification for choice of study . . . 2
Anticipated challenges . . . 2
Learning Goals . . . 3
Links . . . 3
Methods . . . 3
Description of the steps required to reproduce the results . . . 3
Differences from original study . . . 5
Project Progress Check 1 . . . 5
Measure of success . . . 5
Pipeline progress . . . 5
Results . . . 5
Data preparation . . . 5
Key analysis . . . 5
Exploratory analyses . . . 6
Discussion . . . 6
Summary of Reproduction Attempt . . . 6
Commentary . . . 6

### Introduction

[No abstract is needed.] Each reproducibility project will have a straightforward, no frills report of the study and reproducibility results. These reports will be publicly available as supplementary material for the aggregate report(s) of the project as a whole. Also, to maximize project integrity, the intro and methods will be written and critiqued in advance of your

attempt to reproduce the results. Introductions can be just 1-2 paragraphs clarifying the main idea of the original study, the target finding for your reproducibility attempt, and any other essential information. It will NOT have a literature review – that is in the original publication. You can write both the introduction and the methods in past tense.

### Justification for choice of study

The paper by Salomons et al. explains the cognitive development of assistance puppies (aged 8 to 20 weeks) as measured through performance on a biweekly battery of 9 tasks: marker gesture, working memory, working memory with distraction, auditory discrimination, odor discrimination, cylinder (inhibitory control and reversal learning), unsolvable, causal reasoning, and momentary pointing. The results support the multiple intelligences hypothesis and suggest the emergence of distinct cognitive abilities, with clear developmental breakpoints in marker gesture comprehension, inhibitory control, reversal learning, and eye contact occurring before 16 weeks, coinciding with the period of rapid brain development. The goal of this project is to answer if different cognitive skills develop at different rates in assistance puppies. The authors, however, chose to model each task separately and compare developmental rate visually rather than test comparisons directly. Using data exclusively from the 91 longitudinal subjects, I will reproduce Figure 2 and directly test whether age-related improvement differs across tasks using an age $\times$ task interaction. Next, I will simulate additional data from the fitted model to test whether a study of this size (N=91) can reliably detect these differences or is a larger sample (N=1000) is needed for sufficient power.

I am currently working to develop a battery of tests that can be administered to large numbers of dogs, in the hopes of identifying exceptional performers and determining whether strong performance on one task may generalize across different measures of the same ability and/or across cognitive domains. Knowing when different skills emerge will inform when puppies should be tested, and between-task correlations will suggest whether performance may generalize across domains. With the help of citizen science, this method may ultimately identify exceptional dogs.

### Anticipated challenges

As explained above, my plan for this project includes (1) reanalyzing existing data from the 91 longitudinal puppies and (2) simulating additional subjects. My plan for this project will include 6 steps:

1. Reproduce Figure 2 with LOESS curve
2. Reproduce per-task models correct ~ age + (1 | puppy_ID) and correct ~ age + experience + (1 | puppy_ID)
3. Confirmatory mixed-effects logistic model, correct ~ age * task + (1 | puppy_ID) + (1 | puppy_ID:task)

    - Key result is age × task interaction (only including data from the 91 longitudinal subjects, so age and experience will confound)
    - Significant interaction supports multiple intelligences hypothesis
4. Calculate (for how much faster one skill develops relative to another)
5. Data simulation using fitted model (step 3) for N=91 vs N=1000
6. Plot simulated data with LOESS curves like Figure 2

My reproduction will be successful if curves match Figure 2, betas are within +/- 0.01 of the reported, and a find the same best-AIC model per task. My simulation will be successful if slopes show little bias, I can get ~95% CI coverage, and ~80% power for interaction, ideally at N=91.

I anticipate running into a few challenges when attempting to reproduce these results. First, age and experience are highly intertwined since they increase together for all longitudinal subjects. In addition, many of the dogs are siblings, half-siblings, cousins, etc. and this relatedness is not included in the dataset and may be a confound. Finally, of the 11 measures, 10 are binary responses (score out of 4 trials) and one is average looking time, thus I am considering removing the last measure from my simulation for ease. The biggest challenge, however, will be actually completing this project given my minimal programming knowledge and experience.

### Learning Goals

- learn to fit and interpret mixed-effects models for the data
- learn how to write a simulation function that generates correlated data across all 11 measures for the same subject
- learn how to use simulation to test how sample size and testing schedule affect power (directly applies to planning my future project)

### Links

Project repository (on Github): https://github.com/samkefer/salomons2026/tree/main

Original paper (as hosted in your repo): https://github.com/samkefer/salomons2026/blob/main/original_paper/original_paper

## Methods

### Description of the steps required to reproduce the results

1. **Data Preparation.** Before I could begin my reproduction, I first had to prepare the data. The dataset included in the supplements of Salomons et al. does not include subject

ID and thus, you cannot see how an individual dog performs across tests. To fix this error, I used an additional spreadsheet (from a Shared Drive I can still access from my undergrad) to map subject names to each row and assign arbitrary subject ID numbers. After cleaning the xlxs file, each task was exported as a separate CSV file.

2. **Reproduce Figure 2.** Using the cleaned data, I filtered the each CSV to include only longitudinal subjects so I could reproduce Figure 2. For each of the 11 measures, I plotted proportion correct (or mean look time and proportion of trials looked for the unsolvable task) against age in weeks. Points were jittered for individual sessions, a LOESS smoothing curve was applied, and the 11 panels were combined into a single figure using the "patchwork" library.
3. **Reproduce Per-Task Models.** For each binary-outcome task, I fit two mixed-effects logistic regression models, with a random intercept for subject: correct ~ age + (1 | puppy ID) and correct ~ age + experience + (1 | puppy ID), where experience is the number of prior test sessions. The unsolvable task was modeled as proportion of trials looked (logistic) and mean look time (linear). Outcomes were modelled as the number of correct trials out of the total trials in each session. Models were run, and compared, on the longitudinal subjects alone and on entire 113-dog sample. For each sample, the two models were compared using AIC, and the age and experience coefficients of the longitudinal+control models were compared with those reported in the original paper.
4. **Confirmatory Age × Task Models.** The remainder of the analyses were completed using only binary-outcome tasks (i.e., excluding unsolvable task measures) and include only the longitudinal subjects (n=91). To test directly whether cognitive skills develop at different rates, I combined the nine binary-outcome tasks and fitted correct ~ age x task + (1 | puppy ID) + (1 | puppy ID:task) to the 91 longitudinal puppies, with age centred at 8 weeks. The key result is the age × task interaction, wherein a significant interaction indicates that tasks develop at different rates and supports the multiple intelligences hypothesis. Because age and experience are highly correlated in the longitudinal sample, this model cannot disentangle the effects of maturation and testing experience. I also examined the effects of adding a random puppy slope for age, refitting the model with all 113 puppies and task-specific experience effects to see if age can be separated from experience.
5. **Calculate Effect Sizes.** Using the confirmatory model, I extracted per-task effect size (age slope as an odds ratio) and compared each task. Comparing effect sizes serves as a measure of how much faster one skill develops relative to another. Slopes were also expressed as the change in proportion correct per week.
6. **Data Simulation.** Using both the fixed and random-effects confirmatory model, I will simulate trial-level data for each task. Simulations will be run for two different samples: a replication of original sample size (n=91) and a larger sample (n=1000), whose testing schedules will be resampled from the original puppies.
7. **Plot Simulated Data.** Data from both the small and large simulations will be plotted in the same format as Figure 2 using jittered points and LOESS curves for each task. The simulated developmental trajectories can therefore be compared to the actual data and we can also observe the effects of increased sample size on the curves.

### Differences from original study

To complete my analyses, participant ID had to be reconstructed by matching the provided summary data to raw and cleaned DCLB datasets, and a few sessions differed slightly between the three sources. Given the large scope of the original study, I needed to greatly limit the breatdth of my analysis for this project. Unlike the original study, my models only include age and experience, ignoring the potential effects of age and rearing strategy. With this narrowing, I will focus primairly on the longitudnial subjects and will thus not reproduce the longitudinal vs control comparisons of Table 3 or the segmented regression of Table 4. Additionally, the majority of my analyses were not included in the original paper, namely the combined age x task model, random slopes, the model including control puppies, and between-task effect sizes. Finally, Salomons et al. used RStudio 2023.06.1 while my project will use 2026.10.0.

## Project Progress Check 1

### Measure of success

Please describe the outcome measure for the success or failure of your reproduction and how this outcome will be computed.

### Pipeline progress

Earlier in this report, you described the steps necessary to reproduce the key result(s) of this study. Please describe your progress on each of these steps (e.g., data preprocessing, model fitting, model evaluation).

## Results

### Data preparation

Data preparation following the analysis plan.

### Key analysis

The analyses as specified in the analysis plan.

*Side-by-side graph with original graph is ideal here*

### Exploratory analyses

Any follow-up analyses desired (not required).

## Discussion

### Summary of Reproduction Attempt

Open the discussion section with a paragraph summarizing the primary result from the key analysis and assess whether you successfully reproduced it, partially reproduced it, or failed to reproduce it.

### Commentary

Add open-ended commentary (if any) reflecting (a) insights from follow-up exploratory analysis of the dataset, (b) assessment of the meaning of the successful or unsuccessful reproducibility attempt - e.g., for a failure to reproduce the original findings, are the differences between original and present analyses ones that definitely, plausibly, or are unlikely to have been moderators of the result, and (c) discussion of any objections or challenges raised by the current and original authors about the reproducibility attempt (if you contacted them). None of these need to be long.